
Take-Home Assignment: conversation based Agentic AI for Banking CRM

Objective

Design and build an **Agentic AI system** to assist a Relationship Manager (RM) in identifying high-potential customers and generating personalized outreach.

Use Case

A Relationship Manager asks:

“Find high-value customers likely to convert for a personal loan this month and generate personalized WhatsApp messages.”

Expected Capabilities

Your system should:

- Retrieve relevant customer and transaction data
 - Identify high-value customers
 - Estimate likelihood of conversion (rules/ML/heuristics)
 - Recommend suitable products
 - Generate personalized outreach messages based on customer informations
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Expectations

We are **not prescribing architecture**.

You are free to design the system as you see fit.

However, we expect:

- Clear **task decomposition**
 - Effective **tool/API usage** (DBs, functions, services, etc.)
 - Structured **reasoning and execution flow**
 - Proper **state/context handling**
 - Modular and extensible design
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Deliverables

1. GitHub Repository

- Clean, well-structured code
- Clear separation of components (logic, tools, orchestration)

2. README (Mandatory)

Include:

- Architecture diagram
- Execution flow
- Tool design and usage
- Key design decisions
- Trade-offs and limitations
- Setup and run instructions

3. Demo Video (5–10 mins)

- Demo of use cases at least 3
 - Architecture tradeoffs
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Evaluation Criteria

- **System Design** – clarity, modularity, extensibility
 - **Agentic Thinking** – task breakdown, reasoning flow
 - **Tool Usage** – structured, meaningful integration
 - **Output Quality** – relevance, personalization
 - **Clarity** – documentation, trade-offs
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Disqualification

- Hardcoded outputs without reasoning
- No meaningful tool usage
- Poor or missing documentation